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BASIC MATHEMATICS

FORM TWO

Form Two National Assessment (FTNA) 2022 Answer Key and Worked Solutions

Subject	Basic Mathematics
Code	041
Level	Form Two
Year	2022
Exam Board	NECTA
Exam	Form Two National Assessment (FTNA)
Type	Answer key and worked solutions
Questions	10 compulsory questions; 39 answered parts

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: BasicMath-F2-2022.pdf

Question 9(a) supplies only the photograph diagonal. The intended diagonal comparison is shown, with the information needed to guarantee a real rectangular fit explained in its study tip.

Question 1 — Numbers and fractions

1(a)(i). Mwajuma deposits Tsh 360,000. Each withdrawal costs Tsh 1,000. What remains after withdrawing Tsh 106,000?

Step 1: Total account deduction = $106,000 + 1,000 = 107,000$.

Step 2: Balance = $360,000 - 107,000 = 253,000$.

Final Answer: Tsh 253,000.

1(a)(ii). What remains after a further withdrawal of Tsh 50,000?

Step 1: Begin with the previous balance of 253,000.

Step 2: Deduct the second withdrawal and its fee: $253,000 - 50,000 - 1,000 = 202,000$.

Final Answer: Tsh 202,000.

1(b). Rewrite 2.43 as a mixed fraction; only the digit 3 repeats.

Step 1: Let $x = 2.43333\ldots$. Then $100x = 243.333\ldots$ and $10x = 24.333\ldots$.

Step 2: Subtract: $90x = 219$, hence $x = 219/90 = 73/30$.

Step 3: Divide 73 by 30 to obtain $2 \frac{13}{30}$.

Final Answer: $2 \frac{13}{30}$.

Study Tip: The dot is over 3 only: this is $2.4333\ldots$, not the terminating 2.43 or repeating 2.4343...

Section Summary: Keep units consistent, check each condition, and explain the rule used. Drawings need clear labels; area scales use the square of the length scale.

Question 2 — Units and approximation

2(a)(i). Find the total of 30.7 kg, 13.44 kg and 18.26 kg to the nearest whole number.

Step 1: Add exactly: $30.70 + 13.44 + 18.26 = 62.40$ kg.

Step 2: The tenths digit is 4, so round down to 62 kg.

Final Answer: 62 kg.

2(a)(ii). Round dog 30.7 kg to ones, cat 13.44 kg to 1 decimal place, and goat 18.26 kg to 3 significant figures.

Step 1: 30.7 rounds up because $7 \geq 5$; 13.44 rounds down at tenths because the next digit is 4.

Step 2: In 18.26, retain 1,8,2 and inspect the following 6; raise the last retained digit to 3.

Final Answer: Dog: 31 kg; cat: 13.4 kg; goat: 18.3 kg.

2(b)(i). Add 8 km 799 m 400 mm and 5 km 300 m 609 mm.

Step 1: $400 + 609 = 1009 \text{ mm} = 1 \text{ m } 9 \text{ mm}$.

Step 2: $799 + 300 + 1 = 1100 \text{ m} = 1 \text{ km } 100 \text{ m}$.

Step 3: $8 + 5 + 1 = 14 \text{ km}$; retain the 100 m and 9 mm.

Final Answer: 14 km 100 m 9 mm.

2(b)(ii). Convert 14 km 100 m 9 mm into metres.

Step 1: $14 \text{ km} = 14,000 \text{ m}$; $9 \text{ mm} = 0.009 \text{ m}$.

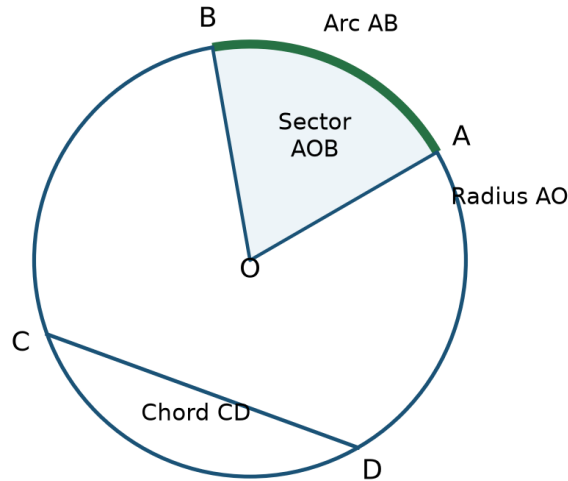
Step 2: Add $14,000 + 100 + 0.009 = 14,100.009 \text{ m}$.

Final Answer: 14,100.009 m.

Section Summary: Keep units consistent, check each condition, and explain the rule used.
Drawings need clear labels; area scales use the square of the length scale.

Question 3 — Circles and area

3(a)(i). Draw a circle centred at O and indicate arc AB.



Step 1: Draw a circle and mark its centre O.

Step 2: Place A and B on the circumference and highlight the curve joining them.

Final Answer: Arc AB is the curved part of the circumference between A and B, shown in green.

3(a)(ii). Indicate chord CD on your circle.

Step 1: Mark C and D as two points on the circle.

Step 2: Join them by a straight line; this is a chord, not the curved arc.

Final Answer: Chord CD is the straight segment joining C and D on the circumference.

3(a)(iii). Indicate sector AOB.

Step 1: Draw radii OA and OB from the centre.

Step 2: Shade the region enclosed by those radii and the arc, rather than the region cut off by a chord.

Final Answer: Sector AOB is the region bounded by OA, OB and arc AB (shaded in the diagram).

3(a)(iv). Indicate radius AO.

Step 1: The centre is O and A lies on the circumference.

Step 2: The straight line from centre to circumference is a radius.

Final Answer: AO is a radius.

3(b)(i). A square carpet has side 14 m. Find the area of the largest circular carpet cut from it.

Step 1: The largest inscribed circle has diameter 14 m and radius 7 m.

Step 2: $A = \pi r^2 = (22/7) \times 7^2 = 154 \text{ m}^2$.

Final Answer: 154 m² (using $\pi = 22/7$).

3(b)(ii). Find the area remaining after cutting the circular carpet.

Step 1: Square area = $14^2 = 196 \text{ m}^2$.

Step 2: Remaining area = $196 - 154 = 42 \text{ m}^2$.

Final Answer: 42 m².

Section Summary: Keep units consistent, check each condition, and explain the rule used. Drawings need clear labels; area scales use the square of the length scale.

Question 4 — Algebra

4(a). Solve $a/2 - b/5 = 1$ and $3b = 24 + a$ using elimination.

Step 1: Multiply the first equation by 10: $5a - 2b = 10$. Rewrite the second as $a - 3b = -24$.

Step 2: Multiply the second by 5: $5a - 15b = -120$. Subtract it from the first: $13b = 130$, so $b = 10$.

Step 3: Then $a = 3(10) - 24 = 6$. Check $6/2 - 10/5 = 1$.

Final Answer: $a = 6$; $b = 10$.

4(b). A book's length exceeds its width by 5 cm and its area is 50 cm². Find its dimensions.

Step 1: Let width be $w > 0$ and length $w + 5$. Then $w(w + 5) = 50$.

Step 2: $w^2 + 5w - 50 = 0$ becomes $(w + 10)(w - 5) = 0$.

Step 3: Reject $w = -10$ because a length is positive; $w = 5$ gives length 10 cm.

Final Answer: Width = 5 cm; length = 10 cm.

Section Summary: Keep units consistent, check each condition, and explain the rule used. Drawings need clear labels; area scales use the square of the length scale.

Question 5 — Ratio and interest

5(a). Asha gets twice Juma's share of Tsh 630,000. How much does Asha receive?

Step 1: The ratio Asha:Juma = 2:1, giving 3 equal parts.

Step 2: Asha receives $(2/3) \times 630,000 = 420,000$.

Final Answer: Tsh 420,000.

5(b)(i). A deposit earns simple interest of Tsh 900,000 in 4 years at 3% per annum. Find the principal.

Step 1: Use $I = Prt/100$; solve for $P = 100I/(rt)$.

Step 2: $P = (100 \times 900,000)/(3 \times 4) = 7,500,000$.

Final Answer: Tsh 7,500,000.

5(b)(ii). Find the amount accumulated after those four years.

Step 1: The amount is the principal plus the simple interest.

Step 2: $A = 7,500,000 + 900,000 = 8,400,000$.

Final Answer: Tsh 8,400,000.

Section Summary: Keep units consistent, check each condition, and explain the rule used. Drawings need clear labels; area scales use the square of the length scale.

Question 6 — Coordinates and scale

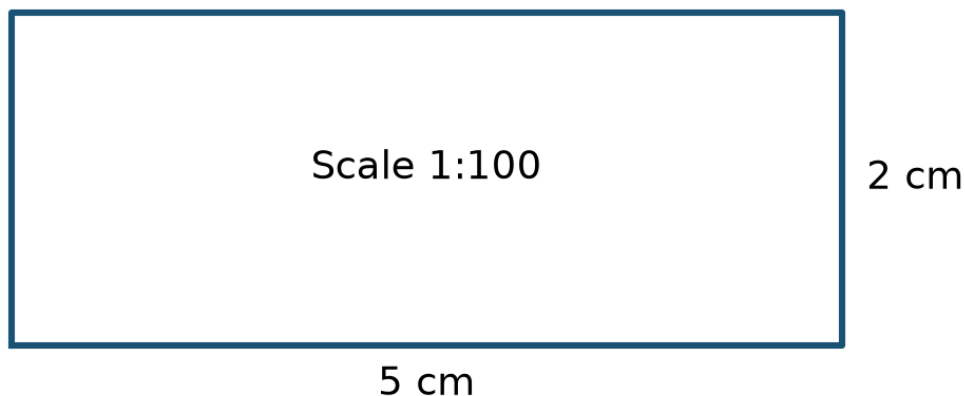
6(a). $y = 3x - p$ passes through (6,10) and (q,22). Find integers p and q .

Step 1: Substitute (6,10): $10 = 18 - p$, so $p = 8$.

Step 2: Substitute (q,22): $22 = 3q - 8$, giving $3q = 30$ and $q = 10$.

Final Answer: $p = 8$; $q = 10$.

6(b)(i). Draw a 500 cm by 200 cm room at scale 1:100.



Step 1: At 1:100, drawing length = actual length $\div$ 100.

Step 2: $500/100 = 5$ cm and $200/100 = 2$ cm. Use a ruler and right angles.

Final Answer: Draw a rectangle of length 5 cm and width 2 cm, labelled with the scale 1:100.

6(b)(ii). Use the scaled diagram to calculate the actual room area.

Step 1: Diagram area = $5 \times 2 = 10$ cm². An area scale is the square of the length scale.

Step 2: Actual area = $10 \times 100^2 = 100,000$ cm²; divide by 10,000 to convert to 10 m².

Final Answer: $100,000$ cm² = 10 m².

Section Summary: Keep units consistent, check each condition, and explain the rule used. Drawings need clear labels; area scales use the square of the length scale.

Question 7 — Radicals and formulae

7(a)(i). $P = \sqrt{2} - 3$ and $Q = \sqrt{2} + 1$. Show $PQ = -1 - 2\sqrt{2}$.

Step 1: Expand: $(\sqrt{2} - 3)(\sqrt{2} + 1) = 2 + \sqrt{2} - 3\sqrt{2} - 3$.

Step 2: Collect ordinary terms and radical terms: $2 - 3 = -1$ and $\sqrt{2} - 3\sqrt{2} = -2\sqrt{2}$.

Final Answer: $PQ = -1 - 2\sqrt{2}$.

7(a)(ii). Show $P/Q = 5 - 4\sqrt{2}$.

Step 1: Rationalise: $(\sqrt{2} - 3)/(\sqrt{2} + 1) \times (\sqrt{2} - 1)/(\sqrt{2} - 1)$.

Step 2: Denominator = $2 - 1 = 1$. Numerator = $2 - \sqrt{2} - 3\sqrt{2} + 3 = 5 - 4\sqrt{2}$.

Final Answer: $P/Q = 5 - 4\sqrt{2}$.

7(b). Express x in terms of p and q if $p = \sqrt{q + x}$; evaluate when $p = 3$, $q = -1$.

Step 1: Square both sides: $p^2 = q + x$. Rearrange to $x = p^2 - q$.

Step 2: Substitute: $x = 3^2 - (-1) = 9 + 1 = 10$. Check $\sqrt{(-1) + 10} = 3$.

Final Answer: $x = p^2 - q$; for $p = 3$ and $q = -1$, $x = 10$.

Section Summary: Keep units consistent, check each condition, and explain the rule used. Drawings need clear labels; area scales use the square of the length scale.

Question 8 — Similarity and congruence

8(a)(i). In the building diagram, $NM \parallel EF$, E, N, G are collinear and F, M, G are collinear. State the similar triangles with reasons.

Step 1: $\angle MNG = \angle FEG$ because $NM \parallel EF$ and NG lies along EG .

Step 2: $\angle MGN = \angle FGE$ is the shared angle at G . Two equal corresponding angles establish similarity.

Final Answer: $\triangle MNG \sim \triangle FEG$ by AA.

8(a)(ii). EN = 6 cm, NG = 18 cm and MN = 12 cm. Find FE.

Step 1: $EG = EN + NG = 6 + 18 = 24$ cm.

Step 2: Corresponding side ratio: $MN/FE = NG/EG$, so $12/FE = 18/24$.

Step 3: $FE = 12 \times 24/18 = 16$ cm.

Final Answer: FE = 16 cm.

8(b)(i). Using the side marks and angles, explain why triangles ABC and PQR are congruent.

Step 1: The double marks show $AC = PQ$; the single marks show $BC = PR$.

Step 2: The included angles $\angle ACB$ and $\angle QPR$ both equal 48° . Two sides and their included angle match.

Final Answer: $\triangle ABC \cong \triangle QRP$ by SAS (equivalently $\triangle CBA \cong \triangle PRQ$).

Study Tip: Match vertices by the side marks: $C \leftrightarrow P$, $A \leftrightarrow Q$ and $B \leftrightarrow R$. Keep that correspondence in the order used to name congruent triangles.

8(b)(ii). In triangle PQR, $P = 48^\circ$ and $R = 72^\circ$. Find angle RQP.

Step 1: Interior angles of a triangle add to 180° .

Step 2: $Q = 180 - 48 - 72 = 60^\circ$.

Final Answer: $\angle RQP = 60^\circ$.

Section Summary: Keep units consistent, check each condition, and explain the rule used. Drawings need clear labels; area scales use the square of the length scale.

Question 9 — Pythagoras and trigonometry

9(a). A photograph has diagonal 7.8 cm; the frame is 6 cm by 5 cm. Show whether it fits.

Step 1: By Pythagoras, frame diagonal = $\sqrt{6^2 + 5^2} = \sqrt{61} \approx 7.810$ cm.

Step 2: $7.8 < 7.810$, so the photograph diagonal does not exceed the frame diagonal.

Final Answer: The examination's diagonal comparison gives "yes": the frame diagonal is approximately 7.81 cm, slightly longer than 7.8 cm.

Study Tip: A diagonal alone does not establish a rectangle's length and width. The intended answer assumes compatible proportions; guaranteeing an actual fit requires both photograph dimensions.

9(b). A square tile has diagonal 8 cm making 45° with a side. Find the side length.

Step 1: The diagonal is the hypotenuse of a right triangle. A side $s = 8 \cos 45^\circ$.

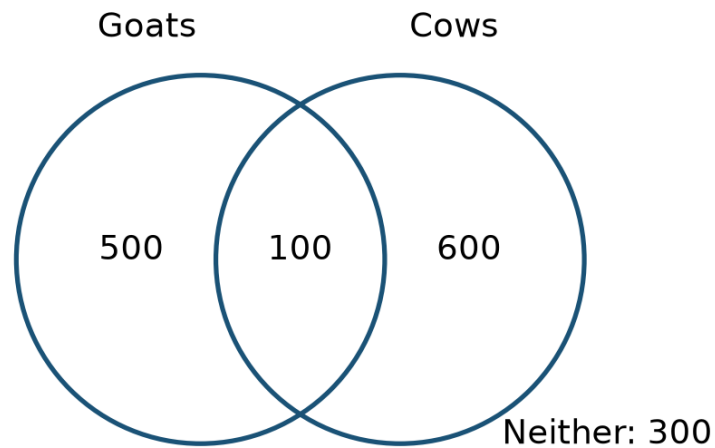
Step 2: Since $\cos 45^\circ = \sqrt{2}/2$, $s = 8 \times \sqrt{2}/2 = 4\sqrt{2}$ cm. Check $s^2 + s^2 = 64$.

Final Answer: $4\sqrt{2}$ cm $\approx$ 5.66 cm.

Section Summary: Keep units consistent, check each condition, and explain the rule used. Drawings need clear labels; area scales use the square of the length scale.

Question 10 — Sets and statistics

10(a)(i). In a village of 1500, 600 keep goats, 700 keep cows and 300 keep neither. Find those keeping both using a Venn diagram.



Step 1: Total in at least one set = $1500 - 300 = 1200$.

Step 2: Goats + cows double-count the overlap: both = $600 + 700 - 1200 = 100$.

Final Answer: 100 villagers.

10(a)(ii). Find the number keeping goats only.

Step 1: The 600 goat keepers include the 100 who also keep cows.

Step 2: Goats only = $600 - 100 = 500$.

Final Answer: 500 villagers.

10(a)(iii). Find the number keeping cows only.

Step 1: The 700 cow keepers include the overlap of 100.

Step 2: Cows only = $700 - 100 = 600$.

Final Answer: 600 villagers.

10(a)(iv). Find the number keeping goats or cows.

Step 1: Or means the union: goats only, both and cows only.

Step 2: $500 + 100 + 600 = 1200$, also $1500 - 300$.

Final Answer: 1200 villagers.

10(b)(i). For mark classes 50-59, 60-69, 70-79, 80-89, 90-99, give the class interval size.

Step 1: Adjacent lower class limits differ by $60 - 50 = 10$.

Step 2: Equivalently, continuous boundaries 49.5 and 59.5 differ by 10, not 9.

Final Answer: 10 marks.

10(b)(ii). Frequencies are 3,21,32,27,17 respectively. Which class has the most students?

Step 1: Compare the five frequencies.

Step 2: The largest frequency is 32, belonging to 70-79.

Final Answer: 70-79 marks, with 32 students.

10(b)(iii). Find the class mark of the highest class interval.

Step 1: Highest class interval means the greatest score interval, 90-99, not the most frequent interval.

Step 2: Class mark = $(90 + 99)/2 = 94.5$.

Final Answer: 94.5 marks.

10(b)(iv). Find how many passed if the pass mark was 70.

Step 1: Include all classes at or above 70: 70-79, 80-89 and 90-99.

Step 2: Passed = $32 + 27 + 17 = 76$.

Final Answer: 76 students.

10(b)(v). Find how many failed with pass mark 70.

Step 1: The failing classes are 50–59 and 60–69.

Step 2: Failed = $3 + 21 = 24$; check passed + failed = $76 + 24 = 100$.

Final Answer: 24 students.

Section Summary: Keep units consistent, check each condition, and explain the rule used.
Drawings need clear labels; area scales use the square of the length scale.